

The State Space and the C₁–C₉ Protocol

Theorem monograph T12 of the chess-dynamics-lab program —
the consolidated protocol

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Keywords: state space, protocol, failure isolation, reproducibility

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Problem statement

Theorems T01-T11 describe the individual layers of the particle model — the board algebra, the kinematics of graphs, mobility, threat fields, energy, the flow, the generation, the memory and the search. But a physics is valid only when all its layers live in one exactly specified state space and are verified by a single protocol. This is the closing link of the program theory: to define the state, to show the structure of its space and to fix the C1-C9 protocol — nine checks, each of which certificate-closes its own theorem.

The principle is inherited from the parent hodge-laboratory program: a statement counts as proved when it is reproduced by a one-command protocol on any machine — from a server to a Termux phone. Therefore the protocol is designed as a set of isolated checks: the failure of one does not block the others, each returns a boolean verdict and details, and the aggregate verdict ALL CHECKS PASSED is attainable only under the full agreement of theory and computation across all nine directions.

The international context: the protocol builds on the classics of chess programming — the community perft references, the endgame tablebase values, the Knuth-Moore bound, Zobrist hashing — and adds its own layer: the three-particle model (threat fields, the Lagrangian, the flow) with its own invariants, above all the equivariance of the fields under the board group of T01.

Theorem

Theorem T12 (the state space and the completeness of the protocol). (i) A position state is the tuple: the board in $\{\text{EMPTY}, \dots, \text{BK}\}^{128}$ in the 0x88 representation, the side to move, the castling rights in 2^4 , the ep square, the halfmove

clock, the move number; the 0x88 encoding gives O(1) boundary and attack tests. (ii) For three-piece endgames the space packs into 22 bits; the realizable states: KRK — 399112, KQK — 368452. (iii) The C1-C9 protocol closes theorems T01-T11: every check is a boolean result with details; a failure is isolated (the exception is caught, the remaining checks continue); the ALL CHECKS PASSED verdict means the simultaneous success of all nine. (iv) The polyglot battery C1-C10 in seven languages prints an identical [PASS]/[FAIL] line format and the final verdict 10/10.

Proof

(i) Specification. The state is realized by the Position structure with a board field of length 128: the index $sq = r \cdot 16 + f$ turns the ~“off-board” test into $sq \& 0x88 \neq 0$ — one bitwise operation instead of two comparisons; the piece offsets are constant arrays that never leave the array bounds. The state round-trip invariant — the FEN cycle: `set_fen(to_fen(P))` restores the position and the hash exactly (verified on all protocol positions and in the pytest tests). The `is_valid` validator checks consistency: exactly two kings, no pawns on the first and last ranks, the consistency of the castling rights and the ep square.

(ii) Reductions of the space. Three compression principles. First — the 22-bit packing of three-piece states (T11): $wk \mid wq \ll 7 \mid bk \ll 14 \mid stm \ll 21$. Second — the symmetry reduction: the group V4 of color-preserving symmetries of T01 acts freely on almost all positions, cutting the equivalence classes roughly fourfold; the bases store orbits, and the protocol verifies the equivariance. Third — irreversibility: the halfmove clock splits the history into continuation trees; in transposition tables a position is distinguishable only together with the clock — otherwise the 50-move rule would break the determinism.

(iii) The mapping of the checks to the theorems is pinned by the table below; every row is a closed loop ~“theorem → check → constants”. Failure isolation is implemented in `run_protocol`: every check runs inside an exception catch; a FAIL returns the details and does not interrupt the rest; the aggregate verdict is the conjunction of all boolean results. The determinism of all checks (fixed seeds, the total move order of T10(iv)) makes the verdict reproducible on any machine — x86 servers and ARM Termux phones alike.

(iv) Cross-language consistency. The polyglot core implements the C1-C10 battery in seven languages with an identical output format: a line ~“[PASS] Cn note” or ~“[FAIL] Cn reason”, ending with ~“verdict: 10/10”. Bit-identical reference constants (perft, orbits, censuses, splitmix64) guarantee that a mismatch of any single backend is caught by comparing the lines. Python, C and JavaScript are verified locally; Rust, Go, Julia and Java are verified in GitHub Actions on every push. The theorem is proved.

Protocol data

The mapping of the protocol to the theorems

Check	Content	Theorem	Key constants
C1	V4/D4 orbits, Burnside lemma	T01	20 and 10 orbits
C2	edge censuses of the move graphs	T02	R448/B280/N168/K210/Q728
C3	mobility: sums and maxima	T03	896/560/336/420/1456; max 27
C4	flow termination t^* , billiard	T06	$t^* = \text{lcm}(W/\text{gcd}(a,W), H/\text{gcd}(b,H))$
C5	perft identities	T08	20/400/8902/197281/4865609
C6	equivariance of threat fields	T04	commutation with D4
C7	Lagrangian energy	T05	$m = 20$; after e4 $m = 30$
C8	retrograde bases + tactics	T10, T11	KRK 16, KQK 10 moves
C9	Zobrist, splitmix64	T09	1562 keys; bound 0.027

Properties of the protocol as a system

Property	Mechanism	Consequence
failure isolation	exception catch in run_protocol	one FAIL hides nothing
reproducibility	determinism T10(iv), fixed seeds	the same verdict on any machine
cross-language reach	polyglot battery C1–C10, 7 languages	the verdict: 10/10 reference line
depth	the -deep mode: perft(5), extended checks	two levels of rigor

Summary of what is proved

We specified the state space of the laboratory, its reductions and invariants, pinned the one-to-one mapping of the checks C1–C9 onto the theorems T01–T11, proved the failure isolation and the reproducibility of the verdict, and the cross-language consistency of the polyglot battery. The program is thereby closed: every theorem has an executable form, and the whole theory has a single verdict verifiable by one command.

Verification

python3 dynamics.py -report — the full protocol with the verdict; python3 dynamics.py -run Ck — an individual check; bash polyglot/run_all.sh — the cross-language battery with the summary table. The full -report takes tens of seconds on a phone (dominated by C5) and seconds on a server.